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### Lab 3: Component Modelling & Architectural Pattern Selection:

System: Internal Microservice Catalog & Health Portal (IMCHP)

Architecture Selection: Microservices Architecture

I choose Microservices Architecture for the Internal Microservice Catalog & Health Portal since the system contains several distinct functions that can be separated into independently manageable service such as service registration, health monitoring, alert notification, API documentation, and dashboard visualization.

#### Reason 1- Independent Service Management:

The health monitoring functionality performs periodic health checks on registered microservices and maintains their health status and availability information. Separating this functionality into a dedicated Health Monitoring Service allows it to operate independently from the dashboard and API documentation components. This is suitable for the system because health monitoring is a continuous backend process.

#### Reason 2- Fault Isolation:

The system must detect three consecutive failed health probes and generate a downtime notification for the DevOps Engineer. Separating the Alert Notification Service from the Health Monitoring Service will provide better fault isolation. A problem in the dashboard or documentation service does not have to stop core health monitoring and alerting functions.

#### Security Advantage:

Microservices provide clear service boundaries through controlled APIs. Access to functions such as service registration and monitoring data can be restricted to authorized users and services, which will reduce unnecessary access between components.

#### Performance Benefit:

The architecture allows resource-intensive services to be scaled independently. In particular, the Health Monitoring Service can handle an increasing number of registered microservices and health probes without requiring the entire portal to be scaled. This supports the requirement for the dependency graph to handle architectures containing up to 200 services smoothly.

#### Conclusion:

Microservices Architecture provides modularity, fault isolation, and independent scaling for the major functions of the Internal Microservice Catalog & Health Portal, which makes it suitable for the system's monitoring, notification, registration, and visualization requirements.